

# Deepwater Well Control: Flow Check & Shut-In Procedure (illustrative)

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## 1. Principles of Deepwater Well Control

In deepwater operations from floating vessels, gas influxes dissolve readily in synthetic oil-based mud (SOBM) until reaching shallow depth or riser entry, making early detection at the mudline essential.

## 2. Warning Signs of Influx

- Sudden drilling break (ROP increase  $> 50\%$ ).
- Increase in flowline return rate.
- Pit gain on active system ( $> 5\text{ bbl}$ ).
- Decrease in pump pressure at constant strokes.

## 3. Soft Shut-In Procedure (Drilling Ahead)

1. Pick up off bottom to space out tool joint across the subsea BOP stack.
2. Stop rotary and shut down mud pumps.
3. Check for flow on trip tank. If flow detected:
  - a. Open remote choke line HCR valve.
  - b. Close upper annular preventer.
  - c. Close remote choke valve.
  - d. Notify Company Man and RTOC immediately.
4. Record Shut-In Drill Pipe Pressure (SIDPP), Shut-In Casing Pressure (SICP), and pit gain.